

ASCII MASTER FLOW – PANEL 1/12: Atom and periodic identity map

NUCLEUS -> protons + neutrons

ATOMIC NUMBER -> protons -> element identity

MASS NUMBER -> protons + neutrons

ISOTOPE / ISOBAR / ISOTONE -> different comparison axes

PERIODIC TABLE -> atomic number + recurring properties

ASCII MASTER FLOW – PANEL 2/12: Bond and force ladder

IONIC -> electron transfer + ion attraction

COVALENT -> shared electron pair

METALLIC -> delocalised electrons

HYDROGEN BOND / VAN DER WAALS -> weaker interparticle forces

TRAP -> do not collapse bond and force categories

ASCII MASTER FLOW – PANEL 3/12: Stoichiometric accounting rail

WRITE SPECIES -> BALANCE EQUATION

CONVERT SUPPLIED QUANTITIES -> MOLES

COMPARE STOICHIOMETRIC REQUIREMENT

IDENTIFY LIMITING REAGENT -> THEORETICAL PRODUCT

ACTUAL / THEORETICAL -> YIELD, only with supplied data

ASCII MASTER FLOW – PANEL 4/12: State and solution matrix

SOLID / LIQUID / GAS -> arrangement + mobility + separation

PHASE CHANGE -> physical identity retained

SOLUTION -> molecular/ionic dispersion

CONCENTRATION -> amount on a stated basis

SOLUBILITY -> equilibrium limit under stated conditions

ASCII MASTER FLOW – PANEL 5/12: Acid base buffer control loop

ACID DONATES H^+ $\leftrightarrow$ BASE ACCEPTS H^+

pH $\rightarrow$ logarithmic activity measure

NEUTRALISATION $\rightarrow$ acidic/basic character reduced

BUFFER $\rightarrow$ consumes limited added acid/base

SALT SOLUTION $\rightarrow$ acidic / basic / neutral depends on ions

ASCII MASTER FLOW – PANEL 6/12: Electrochemical two-cell map

OXIDATION -> anode | REDUCTION -> cathode

GALVANIC CELL -> chemical energy to electrical energy

ELECTROLYTIC CELL -> electrical energy drives chemistry

CORROSION / PLATING / EXTRACTION -> applied routes

SIGN TRAP -> cell type changes signs, not reaction locations

ASCII MASTER FLOW – PANEL 7/12: Kinetics versus equilibrium

RATE AXIS -> collisions + activation energy

CATALYST -> alternative lower-energy pathway

EQUILIBRIUM AXIS -> forward rate equals reverse rate

CATALYST -> faster approach in both directions

NO SHIFT -> equilibrium position unchanged

ASCII MASTER FLOW – PANEL 8/12: Organic and polymer family tree

FUNCTIONAL GROUPS -> alcohol | aldehyde | ketone | acid | amine | ester

ADDITION POLYMER -> no small-molecule elimination

CONDENSATION POLYMER -> small molecule eliminated

THERMOPLASTIC <-> THERMOSET -> reheating boundary

HYDROGEL -> cross-linked hydrophilic network

ASCII MASTER FLOW – PANEL 9/12: Materials structure-property wheel

DIAMOND -> tetrahedral network

GRAPHITE -> layered structure

GRAPHENE / FULLERENE / NANOTUBE -> distinct carbon architectures

ALLOY -> composition + processing alter properties

RULE -> same element can yield different material behaviour

ASCII MASTER FLOW – PANEL 10/12: Fuel conversion fault tree

COMBUSTION -> oxidation + useful heat + possible pollutants

INCOMPLETE COMBUSTION -> different product burden

GASIFICATION -> controlled oxygen/steam conversion

SYNGAS -> carbon monoxide + hydrogen dominant route

OUTCOME FIREWALL -> product is not efficiency or climate proof

MUST REMEMBER: Chemistry fundamentals link atomic structure, mole/stoichiometry, bonding,...

ASCII MASTER FLOW – PANEL 11/12: Environmental compartment map

STRATOSPHERE -> ozone protective role

TROPOSPHERE -> ozone pollutant role

WATER / SOIL -> nutrients, acids, salts, persistent chemicals

EXPOSURE -> concentration + route + duration + conditions

LABEL -> not safety, hazard or regulation evidence

CLOSE DISTINCTION: Atom is not molecule, mole is not mass, ionic is not covalent,...

ASCII MASTER FLOW – PANEL 12/12: PYQ and policy evidence ladder

WATER POLARITY -> many ionic/polar solutes, not every substance

BPA / TRICLOSAN -> presence differs from exposure and regulation

HYDROGEL -> material class differs from universal performance

PRINCIPLE -> PROCESS -> APPLICATION -> INSTITUTION

PAGE / DOCUMENT -> not deployment, outcome, hazard or law

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Balance species, charge and units; state...